

Σ^0 Production in the NuMI Beam at MicroBooNE: Analysis Note

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Abstract

Working note for the MicroBooNE Σ^0 production analysis in the NuMI beam. The selection reconstructs the $\Sigma^0 \rightarrow \Lambda\gamma$, $\Lambda \rightarrow p\pi^-$ chain in $\bar{\nu}_\mu$ charged-current interactions and is finished with a cross-validated gradient BDT. This note records the current state, the planned port into the `xsec_analyzer` framework, and the three pieces of work that follow: a Poisson-thrown fake-data sample, extension from RHC-only to the combined FHC+RHC exposure, and the selection work needed to lift a purity that currently sits near 0.1%.

Contents

1	Status and scope	1
2	Framework port	2
3	Selection	2
3.1	Signal definition	2
3.2	Fiducial volume	3
3.3	Candidate reconstruction	3
3.4	Cut chain	3
3.5	Derived observables	3
3.6	BDT	3
4	Why the purity is low, and what to do about it	3
4.1	The photon candidate: what the data says	4
4.2	Methods worth adopting from the single-photon analyses	5
4.3	Plan: dedicated BDTs in place of one large BDT	6
4.4	Proposed selection work, in order	7
5	Fake data by Poisson throw	8
6	Exposure: RHC now, combined later	9
7	Work plan	9

1 Status and scope

The analysis measures, or more realistically sets a limit on, Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions. The signature is

$$\bar{\nu}_\mu + \text{Ar} \rightarrow \mu^+ + \Sigma^0 + X, \quad \Sigma^0 \rightarrow \Lambda\gamma, \quad \Lambda \rightarrow p\pi^-, \quad (1)$$

so the reconstructed final state is a muon, a proton, a pion and a photon.

The existing selection (`Sigma/pelee/SigmaSelection.C`) runs on standard PeLEE ntuples and reaches, at the operating point $\text{BDTG} > -0.90$ scaled to the full RHC data exposure of 1.1526×10^{21} POT:

Table 1: Current selection performance, RHC only.

Stage	Raw signal	Efficiency [%]	Purity [%]	$S/\sqrt{S+B}$
Preselection (NTracksCut)	650	15.7	0.040	0.038
Cut-based	497	12.0	0.077	0.045
BDTG (current final)	502	12.1	0.108	0.054

Efficiency is respectable; purity is not. A purity of 0.108% means roughly one signal event per thousand selected, and ~ 2640 background events under ~ 2.9 signal events at full RHC POT. **Improving the selection is therefore the priority**, ahead of any normalisation or extraction work: at this purity no downstream treatment can rescue the measurement.

What this analysis produces. Not a differential cross section. At 0.1% purity and 12% efficiency the deliverable is a cross-section *upper limit*. This matters for the framework port (Sec. 2), because `xsec_analyzer` is built end-to-end for differential extraction and contains no limit machinery.

2 Framework port

The selection is being ported into `xsec_analyzer` as a `SelectionBase` subclass, alongside the $\text{CC}1\mu1\pi$ selections. The port buys the framework’s systematic machinery – flux (PPFX multisims), cross-section (GENIE multisims and unisims), detector variations, hadron reinteraction – which is exactly the covariance a limit needs for its background prediction, and which would otherwise have to be built from scratch.

What is ported. Selection, cut flow, truth categorisation, reco and true observables, and the systematic universes at a *single* analysis bin.

What is not. Limit setting, which stays in `CrossSection.C` reading the framework covariance. An unfolded Σ^0 spectrum would be meaningless at this purity, so no multi-bin scheme is planned.

Framework gaps. Three, all small. Nine branches the selection uses are present in the ntuples but unbound in `Branches.hh` (`trk_pid_chipi_v`, `shr_energy_y_v`, `shr_p{x,y,z}_v`, `pi0_energy{1,2}_Y`, `pi0_gammatdot`, `shr_moliere_avg_v`); the fiducial volume is looser than the $\text{CC}\pi$ one and must be set per selection rather than inherited; and the factory needs a registration branch. Details in `SIGMA0_FRAMEWORK_PLAN.md`.

3 Selection

3.1 Signal definition

$\text{nu_pdg} = -14$, $\text{ccnc} = 0$, a Σ^0 (PDG 3212) in the final-state list `mc_pdg`, and the true vertex inside the fiducial volume. Photons are not stored in `mc_pdg`, so the decay is tagged by the Σ^0 itself rather than by its daughters.

3.2 Fiducial volume

$3.00 < x < 253.35$, $-112.53 < y < 114.47$, $3.1 < z < 1033.0$ cm, excluding the dead region $675.1 < z < 775.1$ cm. This is deliberately looser than the CC1 μ 1 π box (10–246, ± 101 , 10–986): the signal is rare enough that acceptance is worth more than vertex quality.

3.3 Candidate reconstruction

- **Muon:** longest track with `trk_score` > 0.8 , start within 20 cm of the vertex, length > 10 cm, and LLR PID > 0.7 .
- **Proton and pion:** assigned jointly by the combined discriminant

$$D = \frac{\text{LLR}}{0.591} + \frac{\chi_p^2 - \chi_\pi^2}{124.5}, \quad (2)$$

with $\text{proton} = \arg \min D$ and $\text{pion} = \arg \max D$. This separates true protons from true pions with AUC 0.873, against 0.838 for χ_p^2 alone and 0.852 for LLR alone. The χ^2 term is dropped per track when undefined ($\sim 30\%$ of tracks).

- **Photon:** the two non-track PFPs closest to the vertex; the closest is taken as the Σ^0 decay photon.

3.4 Cut chain

Nine cumulative stages: `AllEvents`, `FiducialV`, `CosmicVeto` (`topological_score` > 0.20), `NShowers` ($0 < n_{\text{shr}} < 4$), `MuonCandidate`, `ProtonCandidate`, `PionCandidate`, `NTracksCut` ($n_{\text{trk}} < 5$), `BDTG`.

3.5 Derived observables

Λ mass from (p, π) ; Σ^0 mass from (Λ, γ) using the per-PFP shower energy `shr_energy_y_v`; Λ and photon vertex-pointing angles and impact parameters; Λ decay distance; and a π^0 di-photon mass built from the framework π^0 reconstruction for background rejection.

3.6 BDT

Gradient BDT over 49 inputs, evaluated with two-fold cross-validation: folds are trained on even and odd event numbers and each is applied only to the events it did not train on. Per-fold ROC $\simeq 0.81$. The cross-validation is not optional – applying a fold to its own training events would inflate the ROC and the apparent significance.

4 Why the purity is low, and what to do about it

This is the priority. Two diagnostics point at the same place.

The surviving background is π^0 events. RES and DIS together are 76%. Both produce $\pi^0 \rightarrow \gamma\gamma$, which supplies a fake Σ^0 photon, plus charged hadrons that fake the proton and pion tracks. The background is not diffuse: it is one physics process.

The candidate assignment is the weak link. Truth-matched purities of the individual candidates in background events:

Table 2: Surviving background by interaction mode at the final selection.

Mode	Scaled yield	% of background
Resonant (RES)	1381.8	52.3
Deep inelastic (DIS)	635.3	24.0
Quasi-elastic (QE)	363.8	13.8
Meson exchange (MEC)	177.1	6.7
Cosmic	74.6	2.8
Coherent (COH)	8.9	0.3
Total	2643.1	100

Candidate	Correct [%]	Dominant impostor
Muon	81	—
Proton	58	—
Pion	31	another proton (28%)
Photon	29	a proton (18%)

The muon is reliable and the proton is acceptable, but **the pion and photon candidates are wrong roughly 70% of the time**. Since both the Λ and Σ^0 masses are built from them, the two most powerful BDT inputs are being computed from misidentified objects in most background events – and, importantly, we do not know how often in signal events, because the NoCR signal sample’s non-muon truth matching is unreliable.

4.1 The photon candidate: what the data says

This was investigated first, and the result contradicts the obvious expectation, so it is recorded in full.

Composition. Truth-matching the photon candidate in background events (69557 preselected events) gives:

Truth match	Fraction [%]
photon	30.0
unmatched (PDG 0)	26.7
proton	15.2
charged pion	16.3
muon	9.8
electron	1.4

Charged tracks reconstructed as showers – proton, pion, muon together – account for 41%, and a further 27% has no truth match at all.

The signal photon is soft, which inverts the strategy. The Σ^0 – Λ mass splitting is only 77 MeV, so

$$E_\gamma = \frac{m_{\Sigma^0}^2 - m_\Lambda^2}{2m_{\Sigma^0}} = 74.5 \text{ MeV} \quad (3)$$

in the Σ^0 rest frame. The measured mean energy of the photon candidate in signal events is 70 MeV, which matches that prediction and is good evidence that the selection is finding the right object. The background photon candidate averages 140 MeV, because the π^0 photons that dominate it are far harder.

Requiring shower-like quality would therefore remove signal preferentially, which is the opposite of what an earlier draft of this note proposed. A 74 MeV photon is near the reconstruction threshold: it makes few hits and deposits little energy. Measured against true photons in background events – which are π^0 photons and much harder – the signal photon looks *less* shower-like, not more.

Cutting in the correct direction helps, but only before the BDT. Scanning cuts in the soft direction at preselection:

Cut	ϵ_S [%]	Background rejected [%]	FoM gain
$E_\gamma < 120$ MeV	94.5	25.4	$\times 1.092$
$N_{\text{hits}}^Y < 102$	95.7	21.5	$\times 1.079$
$E_\gamma \in [5, 100]$ MeV	70.6	55.2	$\times 1.052$

But `Shower1Energy` is already one of the 49 BDT inputs, so the question is whether the cut adds anything the BDT has not already taken. Applied on top of the BDT operating point it does not:

Selection	S	B	FoM
BDT only	502	20762	3.443
BDT + $E_\gamma < 120$ MeV	482	18663	3.484 ($\times 1.012$)
BDT + $E_\gamma < 100$ MeV	456	17706	3.384 ($\times 0.983$)

Conclusion. The photon candidate cannot be improved by cutting on variables the BDT already has – it is using them close to optimally. An explicit π^0 mass veto is likewise ineffective, rejecting only 9% of background, because it requires *two* reconstructed showers and most background events supply only one. Improvement must come from information the BDT does not yet have.

The one genuinely unused handle. `shr_moliere_avg_v`, the shower Molière angle, is in the ntuples and in the plotting variable list but is *not* among the 49 BDT inputs. It measures lateral spread, which is orthogonal to energy and hit count and is the classic discriminant between an electromagnetic shower and a track stub. Adding it requires re-running the selection to dump it into the training tree, then retraining both folds. That is the concrete next step for the photon candidate, and it should be judged by the FoM gain on top of the BDT, not at preselection.

4.2 Methods worth adopting from the single-photon analyses

The gLEE single-photon search and the NC π^0 technote solve a problem very close to ours: isolate a single photon against a background of π^0 events in which one decay photon was missed. Four of their methods transfer directly.

1. Second-Shower Veto – look for the photon that was *not* reconstructed. This is the most important one. Our π^0 veto requires *both* decay photons to be reconstructed, which is why it fires so rarely and rejects only 9% of background (Sec. 4.1). gLEE instead trains a dedicated Second-Shower Veto (SSV) BDT on evidence that a second photon exists even when Pandora never built a shower from it. Their SSV inputs include:

- the **fraction of unassociated hits in the slice** – hits in the neutrino slice belonging to no reconstructed object, which is exactly what a missed low-energy photon leaves behind;
- the **number of unassociated hits** (used logarithmically);

- a **minimum conversion parameter**, and the implied invariant mass of the candidate photon paired with any second-shower candidate.

Applied to Σ^0 , this attacks the dominant background where it actually lives: our RES and DIS background is 76% of the total and is π^0 -driven, and the failure mode is precisely a missed second photon. This should be tried before any further work on the existing π^0 mass variable.

The ingredients are already in our ntuples, checked directly on the signal file: `slnhits` and `slnunhits` (total and unassociated hits in the slice, whose ratio is gLEE’s fraction variable), `slclustfrac`, `secondshower_Y_nhit` and `secondshower_Y_dot` from the PeLEE second-shower tagger, and `shsubclusters0/1/2` (sub-cluster counts, which the CC1 μ 1 π BDT already uses). None of these is currently read by `SigmaSelection.C`. Adding them is the same mechanical exercise as the Molière angle – bind the branch, add to both variable lists, retrain – but on variables with a far better physics motivation.

2. One BDT per background class, not one BDT for everything. gLEE runs *five* BDTs for its $1\gamma 1p$ channel – Cosmic, ν_e , BNB-Other, NC π^0 and SSV – each trained against one background, with a separate cut on each score. We use a single BDT over 49 inputs trained against all backgrounds at once. Given that our background decomposes cleanly (RES 52%, DIS 24%, QE 14%, MEC 7%, cosmic 3%), a dedicated π^0 -rejection BDT and a separate cosmic BDT would let each learn its own discriminating structure instead of averaging over processes that look nothing alike. This is an architectural change and the single largest departure from the current design.

3. Conversion distance as a first-class variable. gLEE uses the shower conversion distance throughout, in both linear and logarithmic form, as one of its most-shown discriminants. We have `Shower1VtxDist`, which is the distance from the vertex to the shower start and is closely related, but it is one input among 49 rather than a variable the selection is built around. A real photon converts a few centimetres from the vertex; a proton track begins at it. Worth examining separately and in log form, since gLEE’s plots show the discrimination lives at small distances where a linear axis compresses it.

4. Constrain the π^0 rate from data. The NC π^0 technote fits the π^0 normalisation to data and finds GENIE over-predicts non-coherent NC π^0 by 25%: the flat fit gives $N_{\text{non-coh}} = 0.75$ ($\chi^2/\text{ndf} = 1.96/4$), and gLEE separately uses a flat 0.69 or a momentum-dependent $0.92 - 0.80 |\vec{p}_{\text{true}}|$ scaling. Since our background prediction is dominated by the same physics, an unconstrained π^0 normalisation is both a bias and a large systematic. A sideband constraint would reduce both. Note that gLEE deliberately does *not* apply the scaling in its final fit, relying on the covariance matrix instead – that is the more defensible route for us too, and it argues for making sure the π^0 normalisation uncertainty is properly represented in the covariance rather than for rescaling the central value.

What does not transfer. gLEE’s signal is a single photon with no muon, so its ν_e and BNB-Other BDTs target backgrounds we do not have; our muon candidate is already 81% pure and is not where we are losing. Their analysis is also BNB, so the flux and interaction mix differ, and any borrowed *numerical* threshold would have to be re-derived.

4.3 Plan: dedicated BDTs in place of one large BDT

Why the single BDT is the wrong shape. One BDT over 49 inputs is trained against a background that is really four unrelated physics processes – π^0 -producing RES (52%) and DIS (24%), quasi-elastic (14%) and MEC (7%) events with no π^0 at all, and cosmuics (3%). A single discriminant must find one surface separating signal from *all* of them at once, so it learns whatever

those backgrounds have in common and averages away the structure specific to each. gLEE runs five, and their signal is rarer than ours.

The evidence that this is the binding constraint rather than variable quality is that adding variables barely moves the answer: the Molière angle changed the figure of merit by $\times 1.013$, and the ROC plateaus regardless of background size or added inputs. A better-shaped discriminant is a different lever from a better variable.

Proposed structure. Three BDTs, cut independently, replacing the single BDTG:

BDT	Trained against	Fraction of bkg	Discriminating on
π^0 -rejection	RES + DIS with a true π^0	$\sim 76\%$	SSV block, π^0 mass, shower shape
Non- π^0 ν	QE + MEC + COH, no π^0	$\sim 21\%$	track multiplicity, PID, Λ mass
Cosmic	cosmic-tagged events	$\sim 3\%$	topological score, containment, Λ pointing

Each is trained signal-versus-one-class, keeping the same two-fold cross-validation by event-number parity. The truth labels needed to define the classes already exist in the training dump: `Interaction`, `HasPi0`, `CosmicMu` and `Ccnc` are written to `BDTInput.root` as diagnostic branches, so the split costs no new reconstruction, only a change to how `TrainBDT.C` builds its background tree.

How to combine them. Two options, and the choice should be made on measurement rather than taste:

- **Independent cuts**, as gLEE does – one threshold per score, tuned jointly. Transparent, and each cut can be validated in its own sideband.
- **A second-stage combination** – feed the three scores into a small final discriminant. Usually stronger, but harder to validate and easy to overtrain on 650 signal events.

Given our signal statistics, independent cuts are the safer starting point.

The statistics problem this creates, and the answer to it. Splitting the background does not reduce the signal available for training: all 650 signal events are used in every BDT. It does reduce the *background* per training, but from 69557 preselected events, so even the cosmic class retains thousands. The real constraint remains the 650 signal events, which is why the incoming cosmic-overlay signal samples matter more than any of this.

Order of work. Build the π^0 -rejection BDT first and evaluate it against the current single BDT at matched signal efficiency. If it alone does not improve the figure of merit, the split is not the answer and the effort should move to signal statistics instead. Only if it does should the other two follow.

Interaction with the SSV variables. The SSV block belongs naturally in the π^0 -rejection BDT, where its targets are. Adding it to the single BDT first – as has now been done – is worth doing anyway, because it measures how much of the gain comes from the variables and how much from the restructuring: if the SSV variables give little in the single BDT but a lot in a dedicated π^0 BDT, that is direct evidence for the split.

4.4 Proposed selection work, in order

1. **Second-Shower Veto variables** (Sec. 4.2): unassociated-hit fraction and count in the slice. This targets 76% of the background at its actual failure mode and is expected to dominate everything else on this list.

2. **Split the BDT by background class** (Sec. 4.2), starting with a dedicated π^0 -rejection BDT.
3. **Add shr_moliere_avg_v to the BDT.** *Done, and it does almost nothing*, as its separation predicted: 503 signal against 502, purity 0.111% against 0.108%, figure of merit 0.0544 against 0.0537, a gain of $\times 1.013$. Recorded because it closes off the last unused photon-shape variable and supports the argument of Sec. 4.3 that variable quality is no longer the binding constraint.
4. **Improve the pion candidate (31% correct).** The joint discriminant of Eq. 2 is already a real improvement over χ_p^2 alone, but the failure mode is specific: a second proton is taken as the pion 28% of the time. Adding the sign-sensitive information – track length against the proton range curve, or the Λ -mass constraint itself as part of the assignment rather than as a consequence of it – would target that directly.
5. **Reconsider the Λ decay topology.** Λ has $c\tau = 7.89$ cm, so a genuine $\Lambda \rightarrow p\pi^-$ has a displaced vertex. The selection computes `LambdaDecayDist` and the pointing angle but does not cut on them. A displaced-vertex requirement is close to background-free in principle and is the most distinctive feature of the signal.

The analysis is also **signal-statistics limited**: 650 raw signal events at preselection, with a BDT ROC that plateaus regardless of background size or added variables. More signal MC is the largest single lever, and no amount of selection work substitutes for it.

5 Fake data by Poisson throw

The problem. The current “data” sample is the signal file itself (`SigmaSelection.C` maps the same NuWro hyperon ntuple to both sample 0 and sample 4), scaled by POT. That is not a closure test: it contains no background, no statistical fluctuation, and is perfectly correlated with the signal prediction it would be compared against.

The fix. Adopt the same Poisson-throw scheme the CC1 μ 1 π analysis uses (`macros/throw_perrun_*.C`). For each MC event with central-value weight w_{cv} and POT scale $\alpha = \text{POT}_{\text{data}}/\text{POT}_{\text{MC}}$, draw

$$n \sim \text{Poisson}(w_{cv} \alpha) \quad (4)$$

and write the event n times into the fake-data tree with all weights reset to unity. Applied across *all* samples – signal, other hyperons, background, dirt – this produces a fake dataset that

- contains the correct background admixture, which matters enormously here because background outnumbers signal by $\sim 900:1$;
- carries the right statistical fluctuations, so uncertainties can be validated rather than assumed;
- is statistically independent of the prediction, making a genuine closure test possible.

Implementation. A `throw_sigma0.C` modelled on `throw_perrun_w.C`: chain the processed MC, drop the multisim weight branches, `CloneTree(0)` to preserve the Σ^0 observables, throw, and stamp `summed_pot` with the target POT. One seed per sample so the throw is reproducible.

Caveat to keep in view. A Poisson throw from the same MC is a *statistical* closure test, not a model test. It cannot reveal generator mismodelling, because the “data” and the prediction come from the same generator. A genuine model test needs a hyperon sample from a different generator – GENIE against the current NuWro – which does not exist locally. Until it does, the throw validates the statistical and normalisation chain only, and that limitation should be stated wherever the closure is quoted.

6 Exposure: RHC now, combined later

Now: RHC only. Everything is scaled to the full RHC data exposure, $\text{POT}_{\text{data}} = 1.1526 \times 10^{21}$:

Sample	Source	Generated POT
Σ^0 signal / hyperons	NuWro hyperon-enhanced	2.1764×10^{23}
Background	RHC overlay, Runs 1+2+4a+4b+4c	9.94482×10^{21}
Dirt	Run 1 RHC only	4.21274×10^{20}

Two known imperfections: the dirt sample is Run-1 RHC only, scaled up to the full RHC exposure, and the signal sample is generated with no cosmic overlay (NoCR), which is why its non-muon truth matching cannot be trusted.

Later: combined FHC+RHC. The signal is $\bar{\nu}_\mu$ charged-current, so RHC is the sensitive mode – but FHC carries a substantial wrong-sign $\bar{\nu}_\mu$ component (34.0% of the FHC flux, against 53.3% for RHC), and the FHC exposure is 8.857×10^{20} POT. Folding FHC in therefore adds real signal rather than only background, and the $\text{CC}1\mu1\pi$ analysis already has the machinery: per-run POT scaling, a combined `file_properties`, and combined flux normalisation.

Deferred deliberately. The combination is a normalisation exercise, and at 0.1% purity it would add background faster than signal in absolute terms. The selection work of Sec. 4 comes first; the exposure extension is worth doing once the selection is worth extending. When it happens it needs: FHC overlay samples processed through the same selection, an FHC hyperon signal sample (or a justified reuse of the RHC one, given the signal is beam-mode independent at the interaction level), FHC dirt, and a combined-mode flux normalisation following `Flux` in the $\text{CC}\pi$ xsec configs.

7 Work plan

1. **Selection (priority).** Photon-candidate identification, explicit π^0 veto, pion assignment, Λ displaced-vertex requirement. Sec. 4.
2. **Framework port.** Branch bindings, `Sigma0` selection class, factory registration, cut-flow validation against the standalone code before any config work. Sec. 2.
3. **Fake data.** `throw_sigma0.C`, all samples, RHC POT. Sec. 5.
4. **Systematics.** Single-bin `univmake` for the background covariance.
5. **Combined exposure.** FHC+RHC once the selection justifies it. Sec. 6.
6. **Limit.** Outside the framework, reading its covariance.